

New chat



#Outliers

#[1,2,3,4,5, 6,7, 8,9,10,13,40]

#Q1 - 1st quartile(25th percentile) 1,2,3,4,5,

#Q2 - 2nd quartile(6,7)

#Q3 - 3rd/75th percentile ()

Q1 (1st quartile) → 25% of data is below this

Q2 (median) → 50% of data (middle value)

Q3 (3rd quartile) → 75% of data is below this

Together, they split your data into 4 equal parts

Split the data:

Lower half: [1,2,3,4,5,6]

Upper half: [7,8,9,10,13,40]

Q1

#Median of [1,2,3,4,5,6]

#→ middle values = 3 and 4

$Q1 = (3+4)/2 = 3.5$

Q3

#Median of [7,8,9,10,13,40]

#→ middle values = 9 and 10

$Q3 = (9+10)/2 = 9.5$

$Q1 = 3.5$

$Q2 = 6.5$

$Q3 = 9.5$

#sort the values in ascending

```
#median
#q2 --> median --> (6+7)/2 = 6.5
#q1 --> median --> 3
#q3 --> 10

#IQR = Q3 - Q1 = 7 {Interquartile range}
#Q1 - 1.5*IQR ---> LOWERBOUND ---> 3 - (1.5*7) = -7.5
#Q3 + 1.5*IQR ---> upperBOUND ---> 10 + (1.5*7) = 20.5

#40 is not within range ,hence anomaly

#Numbers not within IQR are outliers

help me to understand this
```

You're very close. The main confusion is:

- how quartiles are calculated
- why **40** becomes an outlier
- and one small mistake in your Q1/Q3 values later in the notes.

Let's rebuild it slowly and correctly.

Step 1: Sort the data

Data:

[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 13, 40]

Already sorted.

There are **12 values**.

Step 2: Find the Median (Q2)

Since there are 12 numbers (even count):

Middle positions are:

- 6th value = 6
- 7th value = 7

So:

$$Q2 = \frac{6 + 7}{2} = 6.5$$

✓ Median = **6.5**

Step 3: Split into two halves

For quartiles, split around the median.

Lower half:

[1, 2, 3, 4, 5, 6]

Upper half:

[7, 8, 9, 10, 13, 40]

Step 4: Find Q1

Q1 = median of lower half.

Lower half:

[1, 2, 3, 4, 5, 6]

Middle values:

- 3
- 4

So:

$$Q1 = \frac{3 + 4}{2} = 3.5$$

✓ Q1 = 3.5

Step 5: Find Q3

Upper half:

[7, 8, 9, 10, 13, 40]

Middle values:

- 9
- 10

So:

$$Q3 = \frac{9 + 10}{2} = 9.5$$

✓ Q3 = 9.5

Final Quartiles

Quartile	Value
Q1	3.5
Q2 (Median)	6.5
Q3	9.5

Step 6: Find IQR

IQR = Interquartile Range

Formula:

$$IQR = Q3 - Q1$$

So:

$$IQR = 9.5 - 3.5 = 6$$

✓ IQR = 6

Step 7: Find Outlier Bounds

Outlier rule:

Anything outside:

$$Q1 - 1.5(IQR)$$

or

$$Q3 + 1.5(IQR)$$

is considered an outlier.

Lower Bound

$$3.5 - 1.5(6)$$

$$3.5 - 9 = -5.5$$

✓ Lower bound = -5.5

Upper Bound

$$9.5 + 1.5(6)$$

$$9.5 + 9 = 18.5$$

✓ Upper bound = **18.5**

Step 8: Detect Outliers

Valid range is:

$$[-5.5, 18.5]$$

Check numbers:

- 1 ✓
- 2 ✓
- ...
- 13 ✓
- 40 ✗

Because:

$$40 > 18.5$$

So:

✓ 40 is an outlier

Important Correction in Your Notes

You later wrote:

q1 --> 3
q3 --> 10
IQR = 7

That part is incorrect.

Correct values are:

Wrong	Correct
Q1 = 3	Q1 = 3.5
Q3 = 10	Q3 = 9.5
IQR = 7	IQR = 6

That mistake changed your bounds slightly.

Intuition Behind IQR Outliers

The IQR method looks at the “middle spread” of the data.

Most numbers are clustered between:

3.5 and 9.5

But 40 is extremely far away from the normal pattern.

So it gets flagged as an outlier.

Visual Understanding

1 2 3 4 5 6 7 8 9 10 13	40
-----normal spread-----	^
	outlier

One More Important Concept

“Numbers not within IQR are outliers”

This statement is NOT correct.

Correct statement:

Numbers outside the bounds:

$$Q1 - 1.5(IQR)$$

and

$$Q3 + 1.5(IQR)$$

are considered outliers.

A number can be outside the IQR and still NOT be an outlier.

Example:

- 13 is outside the IQR range (3.5 to 9.5)
- but 13 is still NOT an outlier because it is within the allowed bounds.

That distinction is very important.